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Beyond the city limits: analysing consumption-based emissions and spillover effects in the Swedish urban context

IULIIA RAIKOVA

Supervisors:
Göran Finnveden - KTH Royal Institute of Technology
Henrik Carlsen - SEI

Examiner:
Mattias Höjer - KTH Royal Institute of Technology

Master of Sustainable Urban Planning and Design
KTH Royal Institute of Technology
School of Architecture and the Built Environment
Department of Sustainable Development, Environmental Science and Engineering
SE-100 44 STOCKHOLM

TRITA–ABE–MBT–2634
Stockholm, Sweden 2026
www.kth.se

I Abstract

Global efforts to achieve the Sustainable Development Goals and operate within Planetary Boundaries require comprehensive strategies that account for non-territorial impacts, particularly negative spillover effects driven by consumption in wealthy nations. This highlights a critical disconnect in climate mitigation in high-consumption economies like Sweden mainly focuses on territorial emissions, creating a significant oversight in addressing the true global environmental impact driven by urban consumption patterns.
This thesis evaluates Stockholm, Gothenburg, and Malmö urban strategies and identifies different policy instruments including policy instruments from the Nordic Cooperation report's longlist to pinpoint local policy needs. The analysis revealed a significant implementation gap: municipal strategies remain largely based on territorial accounting, resulting in insufficient integration of policies that effectively address consumption-related emissions and international spillover effects. This thesis concludes with a set of recommendations for local policies aimed at minimizing the impact of Swedish cities on the global climate and promoting positive global effects.

Keywords: consumption-based emissions, Sweden, spillover effects

II Sammanfattning

Globala ansträngningar för att uppnå de Hållbara Utvecklingsmålen och verka inom Planetära Gränser kräver omfattande strategier som beaktar icke-territoriella effekter, särskilt negativa spillover-effekter drivna av konsumtion i välmående nationer. Detta belyser en kritisk koppling som saknas: klimatbegränsningar i högkonsumtionsekonomier som Sverige fokuserar huvudsakligen på territoriella utsläpp, vilket skapar en betydande brist i hanteringen av den verkliga globala miljöpåverkan som drivs av urbana konsumtionsmönster.

Denna avhandling utvärderar de urbana strategierna i Stockholm, Göteborg och Malmö och identifierar olika styrmedel, inklusive de från Nordiska samarbetets långa lista, för att peka ut lokala policybehov. Analysen avslöjade en signifikant implementeringslucka: De kommunala strategierna är i stor utsträckning fortfarande baserade på territoriell redovisning, vilket resulterar i en otillräcklig integration av policys som effektivt hanterar konsumtionsrelaterade utsläpp och internationella spillover-effekter. Denna avhandling avslutas med en uppsättning rekommendationer för lokal politik som syftar till att minimera de svenska städernas påverkan på det globala klimatet och främja positiva globala effekter.

Keywords: konsumtionsbaserade utsläpp, Sverige, spridningseffekter

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IV List of abbreviations

CBA	Consumption-based accounting
CBE	Consumption-based emissions
EE-IO	Environmentally Extended Input-Output
EU	European Union
GHG	Greenhouse gas
GPP	Green Public Procurement
IPCC	Intergovernmental Panel on Climate Change
OECD	Organisation for Economic Co-operation and Development
SDGs	Sustainable development goals
SEI	Stockholm Environment Institute

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1. Introduction

In the context of sustainable development, current strategies and policies increasingly emphasize the importance of addressing spillover effects to achieve the Sustainable Development Goals (SDGs).

The consideration of spillover effects is necessary for effective policy integration, particularly in the context of climate change and sustainable development. Spillover effects refer to the unintended consequences - both positive and negative - of policies that extend beyond their immediate targets, influencing broader economic and social landscapes. Recognizing these effects can enhance the design of interventions, thereby maximizing desired outcomes while minimizing negative impacts (Francetic et al., 2022).

Positive spillovers contribute to the achievement of the SDGs in other countries, while negative spillovers can hinder progress and escalate inequalities. The ability to identify and classify these spillover effects is essential for effective policy-making and international cooperation (Gómez-Paredes et al., 2025).

The consideration of spillover effects in urban strategies and policies is a critical aspect of governance. Spillover effects can significantly impact various sectors, including economics, health, and the environment, influencing outcomes that extend beyond the immediate targets of policy initiatives. “Addressing negative spillover effects in the areas of transformation offers an opportunity not only to tackle them as they are at the moment but also to incorporate their ramifications at international level from an early stage, specifically when new processes are being designed” (Droste et al., 2023).

Sweden frequently appears in international discourse as a leader in environmental policy and sustainable urban development (Global Utmaning,

2017). The country has set ambitious national climate goals, including becoming fossil-free by 2045, and has pioneered environmental policy instruments like the carbon tax (Widerberg et al., 2024). Swedish cities, particularly Stockholm, Malmö, and Gothenburg, have implemented numerous sustainable urban initiatives in areas like energy, transport, waste management, and building design, often showcased internationally (Region Stockholm, 2024). Their success in economic growth from territorial emissions has positioned them as models for urban development worldwide. However, this story of domestic success, based on traditional production-based emissions accounting, conceals a more complex and contradictory reality. In wealthy countries, a significant portion of their environmental impact is offshored. The emissions generated within a city's geographical boundaries represent only a fraction of its total global impact. A more complete assessment requires a shift in perspective to consumption-based emissions (CBE) - the greenhouse gas (GHG) released across the entire global life cycle of goods and services consumed by a city's residents, regardless of where those emissions occur (Hult & Larsson, 2016).

Furthermore, Sweden has been proactive in developing and discussing CBE accounting, acknowledging the impacts beyond its territorial borders. The official Swedish environmental policy objective explicitly states the goal of solving domestic environmental problems "without increasing environmental and health problems outside Sweden's borders" (Engström et al., 2024).

However, this leadership image is complicated by Sweden's high per capita consumption levels and the associated global environmental footprint (Hult, 2017). Studies indicate a significant discrepancy between Sweden's territorial (production-based) emissions, which have decreased, and its CBE, which remain high and are largely generated outside the country (Engström et al., 2024). This suggests that Sweden, despite its domestic progress, may be externalizing a substantial portion of its environmental burden. This potential paradox makes Sweden a particularly relevant and critical case for examining the global spillover effects of ambitious sustainable urban policies implemented within an affluent, high-consumption context. Analyzing Sweden allows for an exploration of whether and how policies designed for local sustainability translate into net positive or negative impacts on a global scale.

The critical role of urban areas in driving global emissions underscores the necessity for comprehensive local climate action, as the Intergovernmental Panel on Climate Change (IPCC) identifies cities as being responsible for

approximately 70% of global carbon dioxide output (IPCC, 2018). For developed metropolitan centers, focusing mostly on emissions generated within city limits fails to capture the full climate footprint. Many developed nations have outsourced high-impact production processes globally, resulting in consumption impacts that often exceed their production-based impacts (Tukker et al., 2020).

Thus, a consumption-based approach is necessary because these developed cities serve as the "final destination" for the consumption of goods and services produced within complex globalized supply chains (Sudmant et al., 2018). By using consumption-based accounting (CBA), policymakers can identify specific leverage points in these supply chains, such as source switching or implementing producer responsibility measures where responsibility for emissions falls on the end consumer rather than the local producer.

Aims and objectives

This thesis aims to identify and analyze policy and implementation gaps between Sweden's national climate goals and specific measures adopted at the municipal level in Stockholm, Gothenburg, and Malmö. The study will use the concept of CBE and spillover effects to critically evaluate both municipal strategies and a comprehensive range of national policy instruments, ultimately developing recommendations for bridging the local and national gaps in addressing global climate impacts.

Research questions:

1. How do academic and grey literature sources on urban sustainability conceptualize and assess spillover effects specifically in relation to the urban contexts of Stockholm, Gothenburg, and Malmö?
2. What policy recommendations can help these cities better manage their global impacts and contribute more effectively to global sustainability within planetary boundaries?

Although Swedish municipalities are often viewed as pioneers in climate governance, existing research mostly focuses on reducing emissions at the territorial level, effectively obscuring the global environmental impacts of local consumption (Axelsson et al., 2021). Despite growing recognition that urban consumption patterns lead to significant negative spillovers in non-territorial jurisdictions (Lager & Benzie, 2022), there remains a documented gap in the practical implementation of municipal strategies to address these indirect emissions (Axelsson et al., 2023).

The thesis is structured as follows: Chapter 2 establishes the theoretical background, defining the core concepts of SDGs, spillover effects and CBE. Chapter 3 details the methodological framework used for the scoping literature review and policy analysis. Chapter 4 presents an overview of Sweden's current environmental goals, analysis of the policies and strategies of Stockholm, Gothenburg, and Malmö. Chapter 5 discusses the broader implications of these findings, synthesizing the theoretical and empirical insights and concludes the thesis with a set of policy recommendations for urban practitioners. Finally, Chapter 6 a summary of key findings and suggestions for future research.

2. Theoretical framework

2.1. Sustainable development goals

This chapter builds upon the SDG framework by introducing key theoretical lenses that inform the thesis: planetary boundaries, global spillover effects, and future scenario planning. The 2030 Agenda for Sustainable Development, with its 17 SDGs, represents a universal call to action to eradicate poverty, safeguard the planet, and ensure prosperity for all by 2030 (Government of Sweden, 2021). These goals are inherently interconnected, and their achievement necessitates addressing complex interdependencies. A critical aspect of this complexity is the recognition that actions undertaken within one nation can significantly influence the capacity of other nations to meet their own SDG targets (SDG Transformation Center, n.d.). Countries operate within a highly interdependent global system where national policies and consumption patterns can generate both positive and negative effects on others' progress towards the SDGs. For instance, the pursuit of economic growth in one region might inadvertently undermine environmental sustainability or social equity in another, thereby hindering the attainment of specific SDGs in the affected region. (SDG Transformation Center, n.d.).

This interconnectedness implies that the SDGs inherently demand a transboundary perspective, moving beyond purely national assessments of progress. The universal and indivisible nature of the SDGs, coupled with the documented impact of international spillover effects, means that a solely domestic focus on SDG achievement is insufficient and potentially misleading if these external impacts are disregarded. This means it is essential, for both research and ethical reasons, to take these spillovers into account. Furthermore, a potential systemic tension exists between high domestic SDG

performance in affluent nations and the externalization of environmental and social costs. Assessments indicate an inverse relationship between domestic SDG progress and the generation of negative spillovers, with high-income countries often demonstrating better domestic SDG performance while simultaneously generating more negative impacts abroad (Berger et al., 2024). This suggests that the prevailing economic models in many high-income countries might enable them to achieve domestic SDG targets by effectively "outsourcing" unsustainable practices, such as resource-intensive extraction, polluting industries, or reliance on supply chains with poor labor standards, to other, frequently lower-income, countries. This dynamic poses a fundamental challenge for global SDG attainment and raises critical questions about the true sustainability of development models and lifestyles in high-income nations.

2.2. Spillover effects

The term "transboundary impacts" is often used synonymously, emphasizing that these effects cross administrative or national borders (Tümer & Tümer, 2024). In the framework of the SDGs, these are sometimes referred to as "SDG spillover effects" specifically denoting the influence one country or region has on another's ability to achieve the SDGs (Gómez-Paredes et al., 2025). These spillovers are transmitted through various interconnected global systems, including international trade (embodied resources and impacts), financial flows, migration, information transfer, and environmental flows (e.g., air and water pollution, shared ecosystems) (Tümer & Tümer, 2024). Recognizing and managing these spillovers is essential for achieving genuine global sustainability and ensuring policy coherence. (SDG Transformation Center, n.d.).

Spillover effects, in a general sense, refer to the impacts – which can be positive or negative – that an event, activity, or policy in one context exerts on outcomes in another, often seemingly unrelated or untargeted, context. These effects arise because systems (economic, social, environmental) are interconnected; actions ripple through these networks. The term often highlights unintended consequences. In the context of globalization, financial integration, and international trade, these effects frequently manifest on a global scale, where events in one country can trigger impacts worldwide (Yang et al., 2021).

Within the field of sustainability, global spillover effects specifically refer to the transboundary impacts that one country's or region's pursuit of its own

development goals (or its general patterns of production and consumption) have on the ability of other countries or regions to achieve sustainable development (Taghvaee et al., 2022). These effects represent the international dimension of sustainability, acknowledging that national actions have global consequences.

Various typologies are used to categorize spillover effects, reflecting the different domains they impact:

Environmental spillovers: These relate to the cross-border movement of environmental pressures or impacts. Common examples include GHG emissions embodied in traded goods (CBE), the depletion of shared or distant resources (like water or forests) driven by import demand, transboundary pollution (acid rain, marine plastic), the export of hazardous waste, and impacts on global biodiversity due to consumption patterns (IDEA Consult, 2024).

Social spillovers: These encompass the effects on social conditions in other countries. Examples include impacts on labour rights and working conditions (fatal accidents, modern slavery) within global supply chains driven by consumer demand elsewhere, effects on health outcomes in neighbouring regions due to disease transmission or lack thereof (herd immunity) or impacts on education and poverty levels linked to international trade or aid policies (IDEA Consult, 2024).

Economic/financial spillovers: These involve the transmission of economic or financial effects across borders. Examples include the impact of trade policies (like tariffs) on global markets, the effects of foreign direct investment on host economies (knowledge transfer, competition effects) international financial flows, the role of tax havens and corporate profit shifting in eroding other countries' tax bases, and the transfer of technology or financial aid (Kotíková et al., 2025).

Behavioral spillovers: Primarily studied in social psychology, this refers to how engaging in one behavior (e.g., recycling) influences the likelihood of engaging in subsequent, related behaviors (e.g., saving energy). These can spill over across contexts (e.g., work to home) or behaviors (Truelove et al., 2014).

Spatial spillovers: Often analyzed in urban and regional economics, these are effects that policies or economic activities in one location have on adjacent or geographically proximate locations. Examples include infrastructure investments benefiting neighboring areas, or pilot cities attracting resources away from their surroundings (Gao & Wang, 2025).

Understanding these different types of spillovers is essential for a comprehensive assessment of the impacts of national or urban policies in a globalized world.

Spillover effects are an inherent consequence of globalization and the increasing economic interdependence among nations. As global supply chains become more complex and international financial flows expand, the potential for actions in one country to have significant and often unacknowledged consequences in others is amplified. High-income countries, due to their typically higher levels of consumption, bear responsibility for trade-related negative spillovers. This reality necessitates that sustainability strategies adopt a global systems perspective, moving beyond purely national accounting to trace impacts along these intricate international chains. The concept of "embodied" spillovers – such as carbon emissions or water use embedded in traded goods and services – fundamentally challenges traditional territorial accounting of environmental impacts and responsibilities. A country might appear to be reducing its domestic environmental footprint while it is merely outsourcing pollution-intensive or resource-intensive production to other nations and importing the finished products. CBA directly addresses this issue by attempting to re-attribute these embodied impacts to the final consumers, thus offering a more accurate depiction of a nation's true global footprint and shifting the discourse on responsibility for global environmental challenges (European Union, n.d.).

Addressing negative international spillover effects invariably requires multilateral solutions and international cooperation. However, establishing and maintaining effective international environmental regimes can be complicated by factors such as ensuring compliance, overcoming free-riding behavior, and addressing power imbalances among states, as highlighted by regime theory in international relations. Thus, even with a growing scientific understanding of spillover effects, translating this knowledge into effective global action remains a significant governance challenge, especially when the required actions impose costs on one nation for the benefit of another or for the protection of the global commons (UN-Habitat, n.d.).

2.3. Defining and measuring consumption-based emissions

CBA offers a fundamentally different perspective on national climate responsibility compared to traditional territorial inventories (Larsson et al., 2022). Instead of focusing on emissions generated within a country's borders, CBA attributes emissions to the final consumption of goods and services by

residents, government, and investments within that country, regardless of where those emissions occurred during the production process (Davis & Caldeira, 2010). Conceptually, a country's CBE can be understood as its territorial emissions minus the emissions embodied in its exports, plus the emissions embodied in its imports (Larsson et al., 2022). This approach aims to capture the total global emissions footprint driven by a nation's demand.

The primary method for calculating CBE is Environmentally Extended Input-Output (EE-IO) analysis. This approach combines economic data from Input-Output tables with environmental data like emissions factors for each sector (Davis & Caldeira, 2010). To handle international trade, sophisticated Multi-Regional Input-Output models are used to trace emissions through global supply chains back to the final consumer in a specific country (Davis & Caldeira, 2010), allowing for the estimation of emissions occurring in China for a product consumed in Sweden, for example (Schmidt et al., 2019).

Calculating CBE is a complex and data-intensive process (Broekhoff, 2019) that has a significant time lag, typically trailing territorial emissions data by two years. Furthermore, the methodologies and underlying databases are continually evolving, which can lead to variability between studies using different models or data vintages (Schmidt et al., 2019). The scope of CBE includes final consumption and investment, while emissions from the production of exports are excluded and attributed to the importing country (Steinbach et al., 2018). Analyses often identify consumption "hotspots" such as food, mobility, and housing as major contributors (Axelsson et al., 2025).

A key aspect of CBE is that the figures are model-dependent estimates and are subject to uncertainties, which contrasts with territorial emissions where direct measurement is sometimes possible (Schmidt et al., 2019). This requires careful interpretation of the results. Ultimately, EE-IO analysis provides a systemic view of the economy, tracing the intricate web of dependencies across industries and borders (Davis & Caldeira, 2010). This shifts the focus from a purely production-focused viewpoint to a more comprehensive, consumption-based understanding of global environmental consequences.

3. Methodology

This chapter details the methodological choices underpinning this academic paper, justifying their selection and outlining their specific application to the study of Sweden's urban sustainability policies and their associated global spillover effects, particularly in the context of urban strategies.
To answer the research questions, the analysis is based on a wide range of sources, including official documents (regional and municipal comprehensive plans, environmental and climate strategies), grey literature (reports from government agencies, independent expert councils, and non-governmental organizations), and peer-reviewed academic research. By examining the formal strategies and the expert critiques surrounding them, this study aims to provide a nuanced assessment of the cities' ambitions, capabilities, and the inherent tensions in their pursuit of becoming globally responsible actors.

3.1 Scoping literature review

A scoping literature review serves as the foundational methodological component of this research. Scoping reviews are a distinct form of research synthesis designed to map existing literature on a broad topic, identifying key concepts, theories, and sources of evidence while clarifying areas where research is underdeveloped (Mina, 2025). Unlike systematic reviews, which aim to answer precise questions via meta-analysis, scoping reviews are particularly suited for exploratory research in emerging fields where terminology is fluid and boundaries are porous characteristics that define the intersection of urban policy and global spillover effects.

To ensure transparency, the review followed the five-stage framework outlined by Arksey and O'Malley (2005) and refined by Levac et al. (2010):

1. Identifying the research questions

The review was guided by the primary query: "How do academic and grey literature sources conceptualize and assess spillover effects specifically in relation to the urban sustainability strategies of Stockholm, Gothenburg, and Malmö?" Sub-questions focused on identifying which metrics (e.g., CBE, planetary boundaries) are currently utilized in Swedish municipal reporting versus academic critiques.

2. Identifying relevant studies

To ensure the search could be replicated, a dual-stream search strategy was employed between January 2025, and November 2025.

- For an academic database search 3 primary databases were queried: Scopus, Web of Science, and Google Scholar. Search strings were constructed using Boolean operators to combine 3 conceptual blocks: geography, context, and mechanism. The following syntax (adapted for each database) was used:

• Block 1 (Geography): ("Sweden" OR "Stockholm" OR "Gothenburg" OR "Malmö")

• AND Block 2 (Context): ("Urban sustainability" OR "Sustainable cities" OR "Urban strategy" OR "Municipal policy")

• AND Block 3 (Mechanism): ("Spillover effects" OR "Consumption-based emissions" OR "Carbon footprint" OR "Telecoupling")

- Grey literature and targeted search: given the practical nature of municipal planning, significant data exists outside academic journals. A targeted "hand-search" was conducted on the websites of key institutions influencing Swedish policy. This included:

• International/EU: OECD Library, European Environment Agency (EEA).

• National: Stockholm Environment Institute (SEI), Mistra Urban Futures, Naturvårdsverket (Swedish EPA).

• Municipal: The official climate strategy documents (Klimatstrategi) and environmental accounts for the three target cities.

3. Study selection (inclusion and exclusion criteria)

Search results were filtered using strict eligibility criteria to ensure relevance to the thesis objectives.

Table 1. Inclusion and exclusion criteria

Criterion	Inclusion	Exclusion
Language	English and Swedish.	Languages other than English or Swedish.
Document Type	Peer-reviewed articles, government reports, white papers, official municipal strategies.	Opinion pieces, purely journalistic articles, blog posts without references.
Relevance	Must explicitly link Swedish urban/national context to external or consumption-based impacts.	Studies focused solely on territorial emissions with no mention of consumption or trade.

4. Charting the data

Selected studies were processed using a standardized data extraction matrix (Excel) to ensure consistent analysis. The charting fields included:

- Bibliographics: author, year, source type (Academic vs. Grey).
- Geographical scope: specific city (e.g., Malmö) vs. national (Sweden).
- Conceptual framework: Does spillover effects or CBE appear in the strategies?
- Identified gaps: authors stated limitations regarding data availability or policy integration.

The initial search yielded 95 results. After screening titles and abstracts, 70 articles were retrieved for full-text assessment. Of these, 32 were excluded as irrelevant to the specific research scope, leaving a final sample of 38 documents for the scoping review.

5. Summarizing and reporting results

The extracted data was synthesized using a thematic analysis approach. Literature was not graded on "quality" (as in a systematic review) but on "relevance".

This scoping process confirmed that while extensive literature exists on Swedish sustainability broadly, there is a gap in operationalizing these concepts at the municipal level.

3.2. Policy analysis

This research employed a qualitative, document-based policy analysis approach, framed by a normative goal to answer research question: What policy recommendations can help Stockholm, Gothenburg and Malmö better manage their global impacts and contribute more effectively to global sustainability within planetary boundaries?

The methodology was structured as a sequential, multi-step process involving data collection, content analysis, policy screening, and synthesis, ensuring the final recommendations are grounded in both current scientific understanding and local governance context. The primary data sources consisted exclusively of secondary, official documents, including Swedish national reports, strategic plans, environmental programs, and sustainability reports published by the city governments of Stockholm, Gothenburg, and Malmö.

The research followed 5 steps to generate the final policy recommendations:

1. Conducting a comprehensive document analysis of the current situation in Sweden regarding consumption-based emissions and spillover effects. This analysis established the key thematic areas and magnitude of the problem that the policy recommendations needed to address, providing the necessary context for the city-level analysis.
2. The “longlist” of 96 policy instruments outlined in the Nordic Council of Ministers report, “Policy Options for Reducing Consumption-Based Emissions” were individually assessed to determine which measures could be incorporated into city-level strategies. Each instrument was assigned a decision (Yes, No, or Partially Yes) with corresponding reasoning regarding its applicability to municipal policy.
3. Focusing on a content analysis of the official strategy and planning documents for Stockholm, Gothenburg, and Malmö.
4. The information gathered in Step 3 was synthesized to identify crucial gaps between the current municipal ambitions (goals and planned actions) and the necessary level of transformative change required, particularly concerning external environmental effects.
5. The final stage involved synthesizing three distinct sets of inputs to formulate the definitive policy recommendations:
   - Applicable policy measures identified and validated in Step 2 (“the longlist”).
   - Explicit goals and tasks related to spillover effects extracted from the municipal strategies in Step 3.

– Recommendations based on the gaps and contextual analysis from Step 4.

The goal of this systematic synthesis was the creation of a set of policy measures designed to operate within planetary boundaries, thereby helping the cities contribute more effectively to global sustainability.

4. Sweden's urban policies and spillover effects

To understand the extent to which Sweden's emissions impact the rest of the world, this chapter begins with an overview of Sweden's current environmental goals and key institutions, as well as the environmental spillovers it has. It then examines how key Swedish urban centers – Stockholm, Gothenburg, and Malmö – are developing and implementing local strategies and measures to transform consumption patterns and reduce the global impact of urban life.

4.1. Sweden's environmental emissions and spillover effects

Sweden has an international reputation as a leader in environmental policy and sustainable development. The nation has established a comprehensive policy architecture aimed at achieving ambitious sustainability goals. These include a target to become fossil-free by 2045 and to achieve 100% renewable energy production, alongside a framework for net-zero GHG emissions by 2045 at the latest, with emissions from Swedish territory to be at least 85% lower than 1990 levels (approximately 71 million tonnes of CO_2 (United Nations Framework Convention on Climate Change, 2021) (Karlsson, 2021). Sweden's commitment to the 2030 Agenda is detailed in its Voluntary National Reviews, which outline progress and challenges in implementing the SDGs (Government of Sweden, 2021). An early national strategy for sustainable development, presented in 2002, emphasized the integration of social, economic, and environmental priorities (Government of Sweden, 2002).

Key institutions play a critical role in guiding and implementing these strategies. The Swedish Environmental Protection Agency (Naturvårdsverket) is a central public agency responsible for issues related to climate, air quality, biodiversity, circular economy, and environmental monitoring, working to achieve Sweden's environmental objectives (Swedish Environmental Protection Agency, n.d.). At the urban level, the Council for Sustainable Cities was established to support municipalities in developing sustainable urban environments, facilitating knowledge exchange, capacity building, and the implementation of the government's policy for sustainable urban development, thereby driving progress on SDG 11 (Sustainable Cities and Communities) (Green Policy Platform, n.d.).

Despite this long-standing commitment and robust institutional framework, recent assessments present a nuanced picture. Even though Sweden's economy is growing with less environmental harm, the country will likely fail to meet its most important environmental goals by 2030. Reviews by the Organisation for Economic Co-operation and Development (OECD), while acknowledging Sweden's strong performance on many environmental indicators, also highlight these implementation gaps and point to recent policy shifts that have created uncertainty regarding the trajectory towards climate targets (OECD, 2025). For instance, a 2023 climate action plan was criticized for projections of short-term emission increases, and some environmental regulations have been relaxed (Swedish Climate Policy Council, 2025). There seems to be a gap between the government's goals and what its current actions are actually achieving and what impact future plans have on those plans. The multi-level governance approach, from national strategies to municipal implementation facilitated by organisations like the Council for Sustainable Cities, is crucial, as many policies that significantly influence global spillovers (e.g., related to consumption, waste management, transport, and energy use) are implemented and managed at the local urban level (Petridou, 2024).

Swedish policymakers and researchers are shifting the focus of environmental responsibility from domestic production to global consumption patterns. While international reporting under the Paris Agreement primarily relies on territorial emissions, Sweden's national "Generational Goal" explicitly incorporates an international perspective, aiming to solve environmental problems without causing increased environmental and health problems outside Sweden's borders (Naturvårdsverket, 2025). This national ambition, coupled with active research and policy discussions surrounding potential consumption-based climate targets reveals a tension (Larsson et al., 2022).

The standard metric for international commitments does not fully encompass the scope of national environmental aspirations or the complexities of Sweden's global impact through consumption. This necessitates the use and further development of complementary metrics like CBE to inform a more comprehensive national climate strategy.

Based on the preliminary statistics for 2024 from the Swedish Environmental Protection Agency, Sweden's total territorial GHG emissions in 2024 amounted to 47.5 million tonnes of CO_2e, excluding emissions and removals from the Land Use, Land-Use Change, and Forestry (LULUCF) sector. This represents a significant decrease of 33% compared to the base year 1990. While emissions continued to decrease between 2022 and 2023, although at a slower pace, falling by just under 2% (or approximately 0.8 million tonnes of CO_2e), in 2024 compared to 2023, emissions rose by approximately 7%, an increase of just over 3 million tons of CO_2e. The primary cause for this increase was the political decision to lower the "reduction obligation" (a law that obligates fuel suppliers in Sweden to mix biofuel into petrol and diesel in order to reduce the GHG emissions (Swedish Energy Agency, n.d.) which led to a higher consumption of fossil fuels, particularly diesel. The impact was most noticeable in the transport and machinery sectors, where emissions increased by 22% and 32% respectively. In contrast, the industrial sector saw a 5% decrease in emissions, largely due to maintenance shutdowns and lower production demand (Naturvårdsverket, 2025).

In a global context, Sweden's direct territorial emissions constitute a small fraction of the total. According to the IPCC's Sixth Assessment Report the average global emissions for the years 2010-2019 was around 56,000 million tonnes annually (IPCC, 2022) and official Sweden's territorial emissions for 2023 were 44.3 million tonnes of CO_2 (Statistics Sweden, 2024) which means Sweden's share is estimated at approximately 0.08%. Within the European Union, according to the Eurostat Sweden's GHG emissions were 41.67 million tonnes of CO_2e, while the total for the entire EU-27 was 3,354.33 million tonnes in 2022 which means Sweden's contribution was 1.24% of the EU's total emissions. (Statistics Sweden, 2024).

To understand the full scope of Sweden's environmental impact requires moving beyond domestic borders to examine environmental and social spillover effects globally. These effects highlight the dissonance between Sweden's high domestic environmental performance and its significant global impact.

Table 2: Sweden's spillover effects.

Category	Spillover effect	Description and impact
Positive spillovers	Green technology export and innovation diffusion	Sweden is a leader in exporting green technologies (Greentech/Cleantech) such as renewable energy systems, waste management solutions, and grid infrastructure (IEA, 2024). By facilitating the adoption of these innovations, Sweden supports other nations - particularly developing economies - in phasing out outdated polluting technologies, which is important for improving resource efficiency and reducing global CO_2 emissions (Han et al., 2025).
	Policy learning and knowledge transfer	Sweden's comprehensive environmental policies and models for sustainable urban development (e.g., in Stockholm, Malmö, Gothenburg) serve as valuable international examples. This transfers a rich knowledge base that can inspire policy changes and capacity building in other countries. (Formas, 2011)
Negative spillovers	Consumption-based emissions (CBE) and carbon footprint	Sweden's consumption-based GHG emissions are significantly higher than its territorial emissions (Schmidt et al., 2019). Over 60% of this footprint is generated abroad, primarily through imports (Larsson et al., 2022).
	Resource depletion and environmental degradation in exporting countries	High import dependency contributes to environmental burdens abroad. Globally, it is estimated that a significant share of deforestation (47%), GHG emissions (33%), and water stress (24%) are driven by international spillovers, primarily through

		trade in commodities (Center for Global Commons, 2023).
	Impacts from raw material mining (e.g., EVs)	The shift to electric vehicles relies on minerals like lithium and cobalt. Mining these materials is linked to significant concerns: e.g. water depletion and severe labour rights abuses, in mining regions. (Luong et al., 2022).

Table 2 illustrates the "sustainability paradox" for Sweden: its high standard of living and advanced domestic environmental policies coexist with, and are partly dependent upon, the externalization of substantial environmental and social costs to other parts of the world. This outsourcing of environmental load allows Sweden to maintain a comparatively green domestic image, but its overall global footprint remains large, challenging the narrative of comprehensive sustainability leadership.

The pursuit of a "green" economy within Sweden, for instance through the rapid electrification of transport, can itself generate new negative spillovers if the full lifecycle impacts and global supply chains of these green technologies are not rigorously and sustainably managed. This creates the risk of "problem shifting" whereby addressing one environmental issue domestically (such as vehicle emissions) potentially exacerbates others elsewhere (such as environmental damage from the extraction of essential minerals for batteries). The complexity and often deliberate opacity of modern global supply chains make it extremely difficult for both Swedish companies and policymakers to fully identify, monitor, and mitigate these negative impacts, particularly those that arise at deeper, less visible levels of the supply chain. (Tillväxtanalys, 2021).

To truly address Sweden's global footprint and operate safely within planetary boundaries, focusing only on national policy is insufficient. The country's overall footprint is determined by its consumption, a significant portion of which occurs in major urban centers. Therefore, to effectively reduce

consumption-based emissions and negative spillover effects, the lens needs to be shifted to the urban level.
The next section will explore the strategic roles and actions of Sweden's largest cities – Stockholm, Gothenburg, and Malmö – in developing and implementing local strategies aimed at transforming consumption patterns and addressing the global impacts of their urban economies. These cities represent laboratories of change, offering illustrative examples of how local governance can contribute to a more sustainable global reality.

4.2. Policy measures for urban strategies

An analysis of policy feasibility in Sweden must be grounded in a thorough understanding of the nation's governance system. Sweden operates under a decentralized model, where power is significantly devolved to the municipal level (Elander et al., 1997). This decentralization places the primary responsibility for planning squarely on the shoulders of the country's 290 municipalities, which have the sole authority to adopt plans and make decisions on their implementation within their geographical boundaries (Boverket, 2023).
This authority is codified in several key legal frameworks.
The Planning and Building Act (PBL) grants municipalities the main responsibility for physical planning and urban development. The Act mandates that each municipality maintain a comprehensive plan (översiktsplan) that outlines the fundamental characteristics of its land and water use, including how it will take into account national and regional goals for sustainable development (Boverket, 2023). The PBL also requires that climate impacts and adaptation are considered in all planning and permitting processes, and it authorizes municipalities to issue permits for construction and development and grants them decisive influence over the siting of renewable energy infrastructure such as wind turbines (Boverket, 2016).
The Environmental Code (Miljöbalken) is Sweden's foundational environmental legislation. This framework law aims to promote sustainable development and is the basis for regulations concerning a wide range of environmental impacts, from air and water pollution to waste management. The Miljöbalken, along with its associated ordinances, provides the legal basis for municipalities to enforce regulations, supervise local environmentally hazardous activities, and implement waste collection and recycling programs (Naturvårdsverket, 2017).

While municipalities possess extensive legal authority, they are not entirely autonomous. The government provides overarching policy frameworks and national objectives, and the county administrative boards (Länsstyrelserna) act as the state's "extended arm" in the regions. These boards are responsible for safeguarding and coordinating state interests in the planning process and can intervene in issues that affect national interests (Riksintressen) or environmental quality standards (Boverket, 2023). The 3 represents Swedish governance hierarchy and environmental responsibilities. This multi-level system of governance is a crucial element to consider when assessing the viability of any new policy measure.

Table 3: Swedish governance hierarchy and environmental responsibilities.

Level	Key actors	Responsibilities
National	Government and Parliament (Supported by Boverket - National Board of Housing)	Sets legislation: PBL and Environmental Code (Miljöbalken) National interests: Defines areas of national interest (Riksintressen). (Ministry of the Environment, 2000). Targets: Establishes national environmental and housing goals (Boverket, 2023).
Regional	County Administrative Board (Länsstyrelsen)	Supervision: Safeguards state interests and health/safety regulations (Boverket, 2016). Coordination: Coordinates planning issues that cross municipal borders, such as infrastructure and regional growth (Boverket, 2016). Intervention: Can repeal municipal plans if they violate national interests or safety standards (Boverket, 2016).
Municipal	The Municipality (Kommunen)	Planning monopoly: Have right to decide on land use (comprehensive plans) (Boverket, 2016). Permits: Grants building permits (Bygglov) (Boverket, 2016). Enforcement: Ensures compliance with the PBL and Environmental Code locally (Boverket, 2016; Ministry of the Environment, 2000).

To understand which policy measures can help to reduce CBE and spillover effects in Swedish cities the Nordic Cooperation's "Policy Options for Reducing Consumption-Based Emissions" report was analyzed (Engström et al., 2024). The core methodology of this document is the "Policy Delphi method" a structured, multi-stage process designed to build consensus and identify divergent expert opinions on complex policy issues. Unlike the traditional Delphi technique which seeks a narrow consensus among a homogeneous group, the Policy Delphi aims to systematically generate and explore a wide range of expert views on the potential consequences and acceptability of various policy options. This process involves multiple rounds of data collection and analysis, allowing participants to reconsider their initial positions considering the collective feedback and evidence (De Loë et al., 2016).

The Nordic report's application of this method to consumption-based policies is particularly valuable as it assesses measures based on three critical criteria: their potential to reduce emissions, their overall feasibility, and their anticipated public acceptance. The "longlist" of policy instruments was introduced in the report. This "longlist" of 96 policies includes various regulatory and economic instruments, such as regulatory standards, consumption taxes, and public procurement.

In order to identify which policy measures from the "longlist" can be applied in the urban context each of 96 policy instruments were analyzed in a way of their applicability (yes, no or partially). The full "longlist" of 96 measures with reasoning of why certain policy instrument can be applied or cannot be applied in cities presented in Appendix (Table 9).

Table 4 illustrates the 35 policy instruments that cities can implement or influence. Table 4 shows that urban authorities can influence public behavior through information campaigns, environmental education, and consumer guides. They can also create local incentives and infrastructure for sustainable consumption with different measures.

Table 4: The longlist consumption-oriented policies in the Nordic countries.

Measure (Engström et al., 2024)	Description (Engström et al., 2024)	Applicable	Reason
Information campaign	Campaigns that inform consumers on products and behavior that cause high emissions of GHG and other air pollutants.	Yes	Cities can run public service campaigns on sustainable consumption, waste reduction, and low-emission lifestyles.

Promotions for vacationers in the home country	Campaigns to increase the status of vacationers in the home country, such as cottage vacations, active vacations or learning vacations such as summer courses.	Yes	Cities can launch tourism campaigns promoting local "staycations" and nearby attractions to reduce emissions from long-distance travel.
Consumer guides and dietary advice	Mandatory information available to consumers on environmental and health effects of food items and diets.	Yes	Local health and environmental departments can create and distribute guides on sustainable food choices.
Environmental education	Develop pedagogical materials that can help teachers increase students' knowledge about food and its environmental impact.	Yes	Cities can develop and integrate educational materials on consumption and climate impact into local school curricula.
Energy and climate advice	A free service that helps consumers become more energy-smart and reduce their energy use, covering topics like heating, efficiency, transport, and costs.	Yes	Cities can have local energy advisory services to help residents reduce their consumption.
Product location at point of sale	Low carbon products are given preferential placement at retail stores, internet sites, etc.	Yes	Cities can negotiate voluntary agreements with local retailers to give preferential placement to low-carbon products.
Rankings and award campaigns	Product manufacturers and/or sellers are given publicly celebrated awards for low carbon performance through government, trade or third-party organisations.	Yes	A city government can create its own awards to recognize and promote local businesses with low-carbon practices.
Approved technology lists	A list of specific low carbon technologies that are given preferential procurement.	Yes	A city's procurement department can create and use lists of approved low-carbon technologies for all municipal purchases.
Advertising regulation	Advertisement restrictions regarding where, when or how products or services with high emission intensities can be marketed.	Yes	Cities can create bylaws to restrict advertising for high-emission goods and services in specific zones, such as near schools or on public transport.

Requirements for improved sustainability	Successively increased demands are placed on food sold in the grocery store to be sustainable, for example through targets/requirements relating to climate, biological diversity, health and social aspects.	Yes	A city can mandate sustainability requirements for food served or sold in its own public institutions like schools, hospitals, and government cafeterias.
Public procurement	Requirements that public agencies must choose low or zero emission products or services when available.	Yes	Cities are major purchasers of goods and services and can set strict requirements for low-emission products.
Reduce emissions from public employee business trips	Reduce the number of long-distance business trips by public employees in the state, regions and municipalities.	Yes	City government can implement policies to limit air travel for its employees and promote virtual meetings.
Linking public procurement to national climate change acts	Legal requirements to consider national climate goals in all public procurement.	Yes	Cities can align their procurement policies with national climate goals through local regulations.
Sector trade body standards	Voluntary product performance standards set by trade organisations to be followed by all outlets in that trade.	Yes	Cities can partner with local trade associations to encourage the adoption of voluntary standards for businesses within city limits.
Supply chain procurement requirements	Consumer-facing outlets establish embodied carbon requirements on intermediate producers.	Yes	A city can require its contractors and suppliers to meet specific embodied carbon standards, influencing the supply chain.
Voluntary agreements by trade organisations	Trade organisations adopt voluntary commitments to reducing embodied and/or usage carbon of products offered to consumers.	Yes	Cities can work with local trade organizations to establish voluntary commitments for their members.
Required recycling	A ban on disposal of products and materials that can be reused or recycled is introduced.	No, but partially yes	This is national level legislation. However, cities play a key role in enforcing waste management

			regulations and may require mandatory recycling of certain materials within national legislation
Sale of leftover food	Offer a range of things based on leftover food from restaurants or commercial kitchens.	Yes	Cities can create policies, partnerships, or local marketplaces to facilitate the sale of leftover food from restaurants.
Prohibition of certain vehicles in urban zones	An environmental zone limits the type of vehicle that may be driven within a specific area.	Yes	Many cities have already implemented this, such as low-emission zones or congestion charging zones.
Shop product choice	Point of sale operators voluntarily restrict products to lower embodied and/or usage products.	Yes	Cities can promote voluntary agreements with local retailers to restrict high-carbon products on their shelves.
Repair checks and repair funds	Consumers get access to repair vouchers when purchasing certain product groups which can be redeemed to reduce the price of repairs or upgrades to the product.	Yes	Cities can create local repair voucher programs or partner with businesses to offer repair services at a reduced cost.
Extension of product lifetime	Enhancement of product lifetime, including removal of planned obsolescence, and ban on disposal of unsold and functional products.	No, but partially yes	Demanding and banning products is national level policy. However, cities can promote and support local repair services, community workshops, and second-hand shops to extend product life.
Information on service life and repairability	Introduce requirements for labelling with expected lifespan of goods within durable goods, linked to a lifespan guarantee.	Yes	Cities can promote transparency and educate consumers about product durability and repair options, although labeling is a national level policy.
Cash compensatory measures	Targeted, compensatory measures proposed to address the regressive nature of environmental taxes (i.e., affecting low-income people more).	Yes	Cities can implement local programs or work with national programs to provide targeted financial assistance to low-income residents affected by environmental taxes.

Remove climate-damaging subsidies for aviation	For example, municipalities and regions that currently support, or own local airports can remove subsidies and deny new construction or expansion of runways.	Yes	Cities can end their financial support or ownership of local airports to reduce aviation's carbon footprint.
EUs sustainable finance taxonomy	A key policy instrument but outside the explicit scope of this mapping.	Yes	Cities can adopt this taxonomy to guide their own investments and procurement decisions.
Enabling recycling	Facilitate for consumers to re-cycle products and materials at end of use.	Yes	Cities are responsible for providing recycling infrastructure.
Enabling product sharing	Creation of the infrastructure to enhance the ability of consumers to share products rather than purchase individual items.	Yes	Cities can support local sharing platforms and libraries of things.
Spaces for sharing	Requirement that developers set aside space for the sharing economy.	Yes	Urban planners can require new developments to include spaces for sharing.
Infrastructure improvements	A key policy instrument but outside the explicit scope of this mapping.	Yes	Cities are responsible for planning and building infrastructure like bike lanes and public transit.
R&D and innovation policy	A key policy instrument but outside the explicit scope of this mapping.	Yes	Cities can fund local R&D in areas like green technology and sustainable urban design.
Work time reduction	Employers and other organizations can choose to facilitate sustainable lifestyles by offering the opportunity to reduce working hours (with a corresponding reduction in wages).	No, but partially yes	This is a company-level or national policy, but it can be applied to city employees.
Targets and accounting of CBE	Set targets for CBE where emissions outside the country's borders are also included.	Yes	Cities can set their own CBE targets and track their progress.
Climate calculations	Tools that calculate individual climate impacts from transport, housing,	Yes	A city can develop and promote tools or apps for

for consumers	consumption and food and provide inspiration for how you can live more sustainably.		residents to calculate their own carbon footprint.
Consumption-based Paris-compliant 1.5 carbon budgets as a climate policy framework	Calculated for the European Union.	Yes	Cities can adopt the principles of the framework.

One example of the application of a policy instrument from the "longlist" is advertising regulation. From summer 2026 fossil fuel advertisements will be banned on city of Stockholm own advertising sites (Stockholms stad, Kommunstyrelsen, 2025). It will make city of Stockholm first municipality in the country to make such a decision

Moreover, some policy instruments can be applied in cities partially. For example, for "required recycling" measure cities play a key role in enforcing waste management regulations and may require mandatory recycling of certain materials within national legislation (Stockholms stad, Kommunfullmäktige, 2024).

The remaining 61 policy instruments from the Nordic report are generally outside the scope of urban governments. They require national or international legislative authority, as well as complex fiscal and regulatory frameworks that are beyond a city's jurisdiction. Cities cannot levy taxes and therefore cannot implement a carbon tax or change VAT differentiation because these are national or regional government responsibilities. Measures like extended producer responsibility, minimum requirements for repairability, and EU Eco-design for sustainable products regulation are set at the national or EU level. Cities must comply with these but cannot create them.

In total, approximately 36 % of policy instruments are applicable to urban strategies, while the other 64 % are not. This clear division highlights that an effective urban climate strategy must focus on the levers of control it possesses local regulations, public engagement, and targeted economic incentives. Cities excel at direct, behavioral, and infrastructural changes. They must rely on national and international governments to implement broader fiscal and legislative changes that affect consumption at a larger scale.

The analysis conducted by Nordic experts concluded that no single policy serves as a "silver bullet" for emissions reduction. Instead, the report emphasizes the necessity of developing "policy packages" – strategic combinations of informational, economic, and regulatory instruments that are mutually reinforcing and collectively more effective than any single measure alone (Engström et al., 2024).
This finding is fundamental for this research, as it shifts the focus from searching for a single ideal solution to understanding how a set of measures can be integrated and adapted to Sweden's urban context. In the next subsection, the urban strategies and associated measures from Stockholm, Gothenburg, and Malmö concerning the reduction of CBE and spillovers will be analyzed. This analysis, which involves identifying the key goals and policy instruments suggested in these strategies, will form the starting point for creating a set of measures and recommendations for each city.

4.3. Urban strategies

In this section plan and strategies of Stockholm, Gothenburg and Malmö will be analyzed to identify the spillover effects considered and how the city will deal with the problem of spillovers in their future strategies.

4.3.1. Stockholm

Stockholm's overarching climate strategy is dual-layered, addressing both territorial (production-based) and consumption-based emissions. The city aims to be completely fossil-free by 2040 and seeks to become "climate positive" by 2030, meaning it intends to remove more emissions than it emits territorially. The territorial goal is complemented by an explicit target for extra-territorial impact management, defined in the Climate Action Plan 2030: GHG emissions shall be halved by 2030 compared with 2019 levels (Stockholm stad, 2024). This consumption goal is mirrored in the Regional Development Plan for the Stockholm region (RUFS 2050), which also sets the long-term target of halving CBE, alongside a target of reducing annual direct emissions to less than 1.5 tonnes per capita. The RUFS 2050 provides long-term guidance for spatial planning and regional growth, anticipating at least 3.4 million inhabitants by 2050 (Region Stockholm, 2017). It has a conflicting dual vision: becoming "Europe's most attractive metropolitan region" (linked to economic expansion and targets for increasing research and development and startups), while simultaneously achieving a "resource-efficient and resilient region without climate-impacting emissions" (Region Stockholm, 2017). This creates

a core tension, as the goal of economic growth risks undermining the ambitious climate targets.

Environment Programme 2030 for the City of Stockholm (2024) acknowledges that indirect emissions originate from sources such as grid-based energy (district heating and electricity) and that they "can occur inside or outside Stockholm". It further clarifies the scope of the CBE goal, stating that emissions originating from consumption by the City and its residents must be "greatly reduced" specifically including those that occur "in Stockholm and in the world at large" (Stockholms stad, 2024). This commitment aligns the city's stated policy responsibility with the full life cycle impact of its consumption.

Accountability for these goals is formalized through the implementation of a carbon emissions budget. This budget limits the city's total emissions to 9 tonnes CO2e between 2024 and 2040. The purpose of this is to act as a "climate compass," guiding decisions and ensuring that the reduction mandates are integrated into long-term planning (Stockholm stad, 2024).

The Environment Programme 2030 designates the City of Stockholm as a "major contracting authority" with the responsibility to "lead the way". The procurements are clearly identified as having significant potential to achieve most of the Programme's environmental goals. A key policy mechanism involves regulating the purchase of specific goods to mitigate indirect environmental effects. The City mandates a specific purchase ban on goods known to contain harmful substances, including PFAS, phthalates, bisphenols, and persistent organic pollutants. Furthermore, the city requires the gradual phasing out of materials such as cast-in-situ rubber in public environments, making environmentally friendly and biodegradable materials the norm for new construction (Stockholm stad, 2024). This strategy acts as a direct policy tool for managing the negative environmental spillovers embedded in global production chains, shifting the responsibility for impact reduction to higher levels, in line with the principles of CBA (Tukker et al., 2020).

Moreover, food consumption is the main driver of household GHG emissions (27% of total) among Swedish residents (Stockholm Environment Institute, 2024). Consequently, Stockholm aims to reduce the climate impact of municipal food procurement through measures such as "increasing the proportion of plant-based items" and systematically reducing food waste. This requires greater knowledge and planning of low-carbon meals in municipal services, including schools, preschools, and elderly care facilities (Stockholms stad, 2024).

Also, the city actively promotes and ensures “circular flows” meaning that resources will circulate through reuse and recycling rather than be burned (Stockholms stad, 2024). This integration of circular economy principles into policy reflects a core component of Nordic approaches to sustainability, which emphasize the maximization of resource use throughout the life cycle of materials to improve the effectiveness of climate policy (Sutton, 2024).

However, the aggressive use of bans and high standards introduces the risk of carbon leakage, where production shifts to unregulated foreign suppliers to avoid costs, potentially displacing emissions globally (Krupnick, 2020). Conversely, Green Public Procurement (GPP) policies also exhibit “significant cross-boundary spillovers” where local GPP adoption can correlate with reduced carbon intensity in neighboring cities through technology and policy diffusion (L. Wang et al., 2025).

A review of RUFS 2050 reveals that the plan lacks any explicit framework for identifying or managing spillover effects. This “spillover blindness” is common in sub-national planning, which often suffers from a “formal accountability gap”, as the responsibility for international impacts rests with national governments (R. E. Engström, 2022). This analysis demonstrates a dissonance within RUFS 2050. It celebrates and promotes the region’s integration into the global economy through trade, investment, and knowledge exchange, yet it fails to systematically account for the corresponding global environmental and social responsibilities. The sustainability it envisions is geographically bounded, creating a blind spot to most of its resource and impact footprint. This “spillover blindness” means that, from a global perspective, the plan’s sustainability claims are incomplete.

Table 5 operationalizes the spillover concept to critically assess the implicit cross-boundary impacts of the plan’s core policies.

Table 5: A spillover effects analysis framework applied to RUFS 2050.

Spillover type	Specific example	Mentioning in RUFS	Potential mitigation strategy
Environmental (negative)	Embodied carbon and biodiversity loss from imported construction materials.	No, beyond the aggregate CBE metric.	Implement embodied carbon limits in building codes; promote use of local, sustainable materials.
Social (negative)	Poor labor standards in the	No.	Use public procurement to

	supply chains of consumer electronics and textiles.		enforce strict social criteria; advocate for stronger EU supply chain legislation.
Technological (positive)	Development of new clean energy or circular economy technologies.	Yes, as a goal for regional competitiveness.	Establish international partnerships to actively disseminate innovations to other cities.
Economic (negative)	Concentration of investment and talent in Stockholm, potentially disadvantageous to other Swedish regions.	No, the focus is on internal regional balance.	Develop inter-regional collaboration strategies to share economic benefits and specialized functions.

Applying a spillover lens reveals the unacknowledged consequences of Stockholm's development model:

Negative environmental spillovers: the plan's goal of building up to 16,000 new homes annually 22 will drive demand for materials like steel and cement, externalizing the environmental impacts of their production (water depletion, habitat loss, pollution) to other countries. These embodied impacts are not captured in the plan's environmental assessments.

Positive technological spillovers: conversely, the target to invest 4.5% of Gross Domestic Product in research and development could generate positive spillovers. Innovations in green technology and sustainable urbanism developed in Stockholm could be disseminated globally, helping other cities in their sustainability transitions (M. Wang et al., 2023).

The plan celebrates Stockholm's global integration through trade and knowledge exchange but fails to account for the corresponding global environmental and social responsibilities.

RUFS 2050 is a guiding framework implemented through a complex web of actors, including municipalities, government agencies, and private companies. This collaborative model fosters flexibility but diffuses responsibility. Official consultation responses highlight concerns about the gap between the plan's ambitions and its implementation. As the County Administrative Board noted a lack of clear analysis on how the plan's land use would affect protected

national interests and how climate goals would be practically achieved (Länsstyrelsen Stockholm, 2017).

4.3.2 Gothenburg

The City of Gothenburg's strategic framework, articulated in its Comprehensive plan and Environmental and climate programme, presents a vision of a sustainable metropolis that is "open to the world". The plan is built on three core strategies to create a "close, cohesive, and robust city" by promoting accessibility, social inclusion, and long-term resilience. A foundational principle of the city's environmental policy is the explicit commitment that "people, animals, and nature in other countries" should not be negatively affected by life in Gothenburg (Göteborgs Stad, 2022). Despite this, the city positions itself as a major industrial and logistics hub, home to the Port of Gothenburg and major oil refineries. These are considered engines of economic growth and facilitate the movement of vast flows of goods and materials, which are the primary contributors to the city's global environmental impact (Stadsbyggnadskontoret, 2009).

The city's Environmental and Climate Programme 2021-2030 sets a clear and ambitious target: to reduce CBE from a 2019 baseline of 9.25 tonnes per capita to 3.3 tonnes per capita by 2030. This requires a steep annual reduction of approximately 7.6%. To achieve this, the city's strategy focuses on leveraging public procurement with a life-cycle perspective and promoting a circular economy to reduce waste and the demand for new production (Klimat 2030, n.d.).

Despite this, Gothenburg faces a severe implementation crisis. The city's independent Gothenburg Climate Council, composed of expert researchers, concluded in its 2023 report that there are "slim prospects" of the city achieving its 2030 climate goals, calling the current pace of reduction "completely inadequate" (Göteborgs Stads klimatråd, 2023). The council's most critical finding relates directly to the consumption-based target. It issued a formal recommendation that the city must develop effective data collection systems, noting that the data quality for tracking CBE is "Poor at the national level, and non-existent at the local level" (Göteborgs Stad, 2021). This reveals that the city has set an ambitious target but it lacks the fundamental data infrastructure to measure progress or effectively manage the policies designed to achieve it. Without local data, the target remains an abstract aspiration rather than a manageable policy objective.

Beyond the carbon footprint of consumption, the city's planning framework largely overlooks the broader category of spillover effects. The city's industrial-logistical complex is a major source of these spillovers. The oil refineries, for example, are responsible for half of the city's territorial emissions, but most of their products are exported, externalizing the climate impact of their use to other regions (Göteborgs Stads klimatråd, 2023). Similarly, the global supply chains flowing through the port are known vectors for negative environmental and social impacts, such as deforestation and poor labour standards (SDG Transformation Center, n.d.).

The city's planning documents lack a formal framework for assessing these spillovers, creating a "blind spot" in its governance. This aligns with critiques from UN-Habitat, which noted that Gothenburg's sustainability work is often organized in "sectoral silos" and lacks a "comprehensive sustainability strategy" to manage such cross-cutting issues (Valencia et.al, 2019).

To bridge the gap between ambition and action, expert analysis points toward several key strategic pathways.

Table 6: Suggested measures for Gothenburg

Measure	Description
Build the "Data-to-action" infrastructure	The city needs to invest in a robust system for tracking CBE at the local level, modelled on established frameworks like the SEI Consumption Compass to make the CBE target manageable
Operationalize the spillover principle	The ethical commitment to do no harm abroad be translated into a formal "Global Responsibility Impact Assessment" for all major industrial and strategic plans. This would make the unseen costs of the city's economic model visible.
Reform governance for systemic change	The city needs to overcome its fragmented and siloed governance structure by creating a high-level, politically supported Climate Change Council with the authority and long-term funding to drive change across all city departments.
Confront economic lock-in	The city needs to use its leverage as a regulator and customer to negotiate binding "Climate Transition Contracts" with its major industrial emitters, establishing clear and verifiable pathways to decarbonization.

These measures can be a starting point towards strategy objectives. Gothenburg can move from having a climate strategy that is progressive in principle to one that is effective in practice, accounting for its global

responsibilities and closing the critical gap between its ambitious vision and its current reality.

4.3.3 Malmö

The City of Malmö has positioned itself as a European leader in sustainable urban development, distinguished by a sophisticated and integrated approach to its climate responsibilities (Galleta et al., 2021). Unlike many municipalities that focus only on emissions produced within their own borders, Malmö has formally adopted a strategy that confronts its global footprint. This is achieved by targeting CBE and by implementing policies that implicitly manage broader environmental and social "spillover effects" (City of Malmö, 2021). Malmö's climate strategy is built on a cohesive framework that links long-term spatial planning with specific, measurable environmental targets.

The Comprehensive Plan (Översiktsplan 2023) sets the long-term vision for a sustainable, growing city. Its core spatial strategy is to create a "close, dense, green, and mixed-use city" a physical form designed to reduce transport needs, improve resource efficiency, and minimize land use. The plan acknowledges the need to reduce social and economic divisions, establishing a commitment to a just and equitable transition (City of Malmö, 2023).

The Environmental Programme 2021-2030 is the city's primary governing document for climate action, legally binding on all municipal authorities. It is fully aligned with the Paris Agreement and Agenda 2030 and its vision states that "In Malmö, we take responsibility locally and are a role model globally" a principle made concrete through the formal inclusion of a target for CBE (City of Malmö, 2021 c) (Stockholm City Board, 2020).

Climate Transition Malmö operational framework translates goals into practice. Reinforced by Malmö's participation in the EU's "100 Climate-Neutral Cities Mission", this framework organizes the work into key areas, including "climate-smart consumption". A central tool is the "Climate Contract Malmö", which has mobilized nearly 250 actors from business, academia, and civil society to commit to concrete climate actions (City of Malmö, 2024).

Malmö's approach to CBE is a standout feature of its climate policy, characterized by a clear target, a sophisticated measurement tool, and a dedicated action plan. The city's Environmental Programme sets a highly ambitious goal to halve per capita CBE by 2030, reducing them from a 2019 baseline of 6.2 tonnes of CO_2e per person to 3.1 tonnes. This target aims for a footprint significantly below the current Swedish average of around 8-9 tonnes (Wamsler et al., 2021). To manage this, Malmö utilizes the SEI's "Consumption

Compass", a tool that provides high-resolution, postcode-level data on household emissions and reveals critical insights (City of Malmö, n.d.). This data exposes a stark "climate inequality" within the city, with footprints in affluent areas reaching nearly 9,98 metric tonnes per person per year - almost three times higher than in low-income areas, which average around 3,72 tonnes (Stockholm Environment Institute, n.d.). Air travel is a key driver of this inequality, with emissions from flights being ten times higher in some wealthy neighbourhoods compared to poorer ones. In response, the city has developed a "Roadmap for Climate-Smart Consumption" which aims to stimulate and drive change in key areas such as food, goods, and mobility by promoting plant-based diets, encouraging reuse and repair, and shifting travel from air to rail (City of Malmö A, n.d.).

While the term "spillover" is not frequently used in its official documents, Malmö's policies are functionally designed to manage the cross-boundary impacts of its economic activity. The CBE target is the city's primary tool for managing negative environmental spillovers, as it takes into account the climate impact of imported goods and services (City of Malmö, 2021 c), an approach that aligns with Sweden's national "generational goal" to solve environmental problems without exporting them to other countries (County Administrative Board of Skåne, n.d.). Furthermore, the city's "Roadmap for a Circular Economy" mitigates the spillovers of resource extraction by promoting repair, reuse, and industrial symbiosis. This strategy reduces demand for virgin materials, thereby reducing associated environmental degradation, such as deforestation and water pollution, which often occurs in other countries (City of Malmö. D, n.d.).

Recognizing the limits of its direct authority over private consumption and global supply chains, Malmö employs an innovative governance model based on collaboration and co-creation. The city acts as a coordinator, collaborating with universities on applied research, such as the "Neutralpositive Malmö" project focused on a just transition, and mobilizing the private sector through initiatives like LFM30 for a climate-neutral construction sector (Olsson, 2023). Recognizing the importance of citizen engagement, Malmö has also implemented innovative engagement models, such as the demographically representative Climate Assembly ("Klimatrådslag"), where citizens discuss and develop policy recommendations for the city.

Despite this innovative model, significant challenges remain. The vast inequality in consumption footprints creates a major "just transition dilemma": how to curb the high-impact consumption of the affluent without

penalizing low-income communities that already have low emissions (Stockholm Environment Institute, n.d.).

Moreover, the reliance on voluntary, collaborative action may be insufficient to drive the rapid, systemic change needed to meet the 2030 targets, particularly in sectors like transport where car ownership remains high (Climate-KIC, 2025).

Based on this analysis, the following recommendations are proposed to strengthen Malmö's approach, improve the integration of its policies and maximize its impact.

Table 7: Recommendations for Malmö

Measure	Description
Develop a more integrated spillover framework	The city should pilot a project to develop a set of "spillover indicators" for its public procurement. Using data from the Consumption Compass to identify high-impact consumption categories (e.g., IT equipment, textiles, certain food products), the city could develop procurement criteria that specifically target the primary upstream spillovers associated with these goods, such as water footprint, biodiversity impact, and documented labor conditions. This would create a direct, data-driven link between the city's consumption footprint and its procurement actions, making the management of social and broader environmental spillovers as explicit as its management of carbon.
Strengthen levers for a just consumption transition	The data from the Consumption Compass clearly shows that a uniform approach to reducing consumption would be inequitable. The city must move beyond general awareness campaigns to design and pilot differentiated policies that reflect the socio-economic realities of its diverse communities, drawing on the research from the "Neutralpositive Malmö" project. For high-income and high-emissions areas, the city should develop targeted interventions aimed at high-impact behaviours. This could include partnerships with travel agencies to promote sustainable holiday packages, collaborations with retailers in affluent areas to showcase repair and reuse services, and developing hyper-local communication that speaks to the values and norms of these communities. At the same time for low-income and low-emission areas, the focus should be on "climate-positive"

	investments that improve quality of life and prevent future emissions growth. This includes prioritizing these areas for investments in high-quality public transport, creating community-based circular economy infrastructure (e.g., tool libraries, repair cafes, community kitchens), and ensuring that energy efficiency retrofits in social housing lead to lower energy bills for residents, not just reduced emissions for the city.
Leverage the coordinator role to advocate for systemic change	Malmö's most significant long-term impact may come from its ability to influence policy beyond its own borders. The city should strategically leverage its leadership position within influential networks – such as the EU's 100 Climate-Neutral Cities Mission, Viable Cities, and ICLEI – to advocate for the systemic changes at national and EU levels that are necessary to create a supportive environment for its local efforts. The city should form a dedicated policy advocacy group, drawing on its network of partners in the Climate Contract, to develop and promote a specific policy agenda. This agenda should focus on an "enabling framework for sustainable consumption" calling for national and EU-level policies such as: mandatory carbon labeling for consumer products, stronger "right-to-repair" legislation to extend product lifespans, fiscal reforms that shift taxes from labor to resource consumption, and the adoption of national consumption-based emissions targets. By using its own experience as a case study, Malmö can provide credible, practice-based evidence for why such systemic changes are essential for all cities to succeed in their climate transitions.

The City of Malmö has developed one of the world's most advanced municipal strategies for addressing global climate impacts. Its integrated policy architecture, data-driven CBE target, and functional approach to managing spillover effects set a new standard for urban climate responsibility. However, the city's success is constrained by a fundamental governance gap between local goals and global systems. To accelerate the transition, Malmö needs to continue to refine its policies to ensure a just transition, strengthen its co-governance model, and strategically leverage its leadership role to advance the national and international policy changes needed to create a sustainable global economy.

5. Discussion

This chapter combines the theoretical underpinnings, methodological approach, and empirical findings of the thesis to get a deeper understanding of how Swedish cities are navigating the complex challenge of their global environmental footprints. It critically assesses the role of Stockholm, Gothenburg, and Malmö in global sustainability, balancing their celebrated domestic successes with their broader international responsibilities. Finally, it combines the findings of the research into a set of recommendations in addressing CBE and spillover effects.

In order to answer research question: how do academic and grey literature sources on urban sustainability conceptualize and assess spillover effects specifically in relation to the urban contexts of Stockholm, Gothenburg, and Malmö, the theoretical framework of this thesis grounded in the concepts of SDGs and spillover effects, assume that for urban development to be sustainable, The analysis of cities official plans and strategies reveal a clear and growing recognition that the traditional, production-based accounting of GHG is insufficient. By incorporating CBE into their climate targets, these cities are attempting to align their local actions with the global imperative of the climate change planetary boundary.

The adoption of CBE targets by all three cities marks a significant evolution in urban climate governance. It represents a move away from a territorial view of responsibility towards a more globally conscious one, aligning with Sweden's national "generational goal" of not exporting environmental problems. This shift is not merely technical; it is an ethical and political statement.

While all three cities acknowledge the importance of their global impact, their approaches to integrating CBE emissions and spillover effects into their core

planning frameworks reveal distinct levels of maturity, ambition, and operational capacity.

The table 8 illustrates a comparative analysis of Stockholm, Gothenburg, and Malmö address climate change mitigation, specifically focusing on CBE. The comparison highlights varying levels of ambition, data infrastructure, and strategy across the cities in four key areas.

Table 8: Comparison of urban strategies of cities.

Feature	Stockholm	Gothenburg	Malmö
CBE Target	Half by 2030 (vs. 2014) (Region Stockholm, 2017)	3.3 tonnes/capita by 2030 (Göteborgs Stad, 2022)	3.1 tonnes/capita by 2030 (Wamsler et al., 2021)
Data & Monitoring	High-level regional goal; specific monitoring tools not central to plan.	Target set, but local data infrastructure is "non-existent" (Göteborgs Stads klimatråd, 2023).	Actively uses SEI Consumption Compass for data-driven policy (Axelsson et al., 2025)
Action Plan	General strategies (procurement, circularity) (Region Stockholm, 2025).	Ambitious procurement goals; circular economy focus (Klimat 2030, n.d.).	Dedicated roadmaps for consumption and circular economy (Malmö stad, n.d.).
Spillover Framework	"Spillover blindness"; focus on internal regional balance (R. E. Engström, 2022).	Strong ethical principle, but practical "blind spot" for industrial spillovers (Göteborgs Stad, 2021).	Functional approach using CBE, circularity, and procurement to manage different spillovers (Malmö stad, 2021).

The table 8 reveals that both Stockholm and Gothenburg suffer from "spillover blindness" meaning their plans often neglect the external, global emissions embedded in local consumption. Malmö is the only city shown to have a functional approach explicitly designed to manage these spillover effects through its dedicated roadmaps for CBE, circularity, and procurement.

While all three cities have targets, Malmö's framework is currently the most robust and actionable because it effectively pairs ambitious goals with strong data and an integrated mechanism to account for the wider global effects of local consumption.

To answer for the research question about what policy recommendations can help Stockholm, Gothenburg and Malmö better manage their global impacts and contribute more effectively to global sustainability within planetary boundaries this research has evaluated the cities' ambitions, methodologies, and governance models in confronting their global environmental and social footprints by synthesizing insights from official documents, grey literature, and academic research.

While Stockholm, Gothenburg and Malmö serve as necessary laboratories for complex data modeling and market regulation, the implications of CBE are geographically agnostic (Axelsson et al., 2024). Affluent small-town residents often possess carbon footprints equal to or higher than their metropolitan counterparts due to larger housing, car dependence, and similar import consumption. The relevance for smaller cities is grounded in three emerging realities: circular economy strategies offer vital economic relief for budget-constrained councils (Öberg, & Lundbohm, 2025), immediate urban planning is required to prevent infrastructural "lock-in" that would otherwise guarantee failure to meet 2045 net-zero targets (Moodie et al., 2021), and CBA enables climate justice by highlighting disparities between affluent emitters and vulnerable groups (Azar & Johansson, 2024). Consequently, CBE strategies are not merely a "big city luxury" but a fundamental component of any credible municipal strategy, differing only in method - where big cities shape global markets and small cities build local resilience - to ensure Sweden's national compliance.

In the research 5 principal findings that characterize the current state of practice in Sweden's leading cities were identified:

1. Swedish cities, and Malmö in particular, are at the global forefront of urban climate policy by formally recognizing and setting quantified targets for CBE. This marks a crucial paradigm shift from a purely territorial view of climate responsibility to one that acknowledges a city's role in a globalized system.
2. Despite shared ambitions, there is a significant gap between the cities in their operational capacity. Malmö's data-driven approach, utilizing the SEI Consumption Compass, stands in stark contrast to Gothenburg's critical lack of local data infrastructure for monitoring its own target. This highlights that political will must be matched by investment in institutional and technical capabilities.
3. While the cities are beginning to address the climate dimension of spillovers through CBE targets, their strategies largely fail to account for

the broader spectrum of environmental (e.g., biodiversity, water) and social (e.g., labour rights) spillovers. These impacts remain "out of sight" in official planning documents, reflecting a systemic governance challenge for sub-national authorities.

4. A fundamental tension persists within the cities' core strategies between their objectives for conventional economic and population growth and their goals for achieving absolute reductions in CBE and resource use. The plans generally lack a framework for reconciling these potentially conflicting imperatives.
5. Faced with limited jurisdictional power over global supply chains and private consumption, the cities have adopted innovative but potentially insufficient governance models based on orchestration, collaboration, and voluntary action. While these models are crucial for building societal consensus, their ability to drive the required pace and scale of systemic change remains a critical uncertainty.

Based on the findings, the following recommendations are proposed for Swedish cities to more effectively manage their global footprints:

- Build the foundational data infrastructure. All municipalities need to measure and monitor CBE as critical infrastructure. This involves investing in and systematically integrating tools into the core of the municipal environmental monitoring and reporting cycle, making it a basis for evidence-based policy and public accountability.
- Develop an integrated spillover assessment framework. Cities should not focus only on carbon but try to integrate a broader "Global Responsibility Impact Assessment" into their planning processes for major urban development and infrastructure projects. This framework should systematically screen for potential negative spillovers in high-impact sectors like construction and food, considering metrics for water use, biodiversity, and social risks in supply chains.
- Design policies for a just consumption transition. Cities should design targeted interventions that aim to reduce the high-impact consumption of affluent groups (e.g., through promoting alternatives to air travel) while simultaneously making climate-positive investments in low-income communities that enhance well-being and social equity (e.g., improving public transport and access to green spaces).
- Leverage municipal power to advocate for systemic change. Recognizing the limits of local authority, cities need to embrace their role as powerful advocates. They should use their collective voice within

national and EU networks (such as Viable ities and the EU's 100 Climate-Neutral Cities) to push for higher-level regulatory and fiscal policies - such as carbon pricing on imports and extended producer responsibility.

Figure 1 illustrates a vision for a cohesive urban strategy composed of three integrated components: (1) a synthesis of the 35 policy instruments analyzed in Section 4.2 (from the Nordic Council of Ministers report); (2) the urban strategies and plans discussed in Section 4.3; and (3) the recommendations for Swedish cities proposed above. Integrating these elements offers a robust pathway for cities to more effectively manage their global footprints, CBE and spillover effects in an urban context.

Figure 1. A vision for a cohesive urban strategy.



6. Conclusions

This thesis researches a connection between concepts in global sustainability governance: the achievement of the SDGs, the accountability for CBE and the mitigation of pervasive spillover effects. It aims to show that these elements are not simply adjacent but are tightly interconnected and mutually influencing. A failure to address CBE translates directly into escalating global footprints and detrimental spillovers, threatening the Earth's life support systems and making local SDG progress unsustainable.

The main finding of this research is the urgent need for a paradigm shift away from the narrow confines of territorial emissions accounting. The current reliance on production-based metrics fundamentally obscures the true burden of environmental impacts on national and urban economies, shifting responsibility and environmental risk to distant shores. For countries and cities that are the main drivers of global consumption, it is necessary to adopt a truly global perspective and institutionalize the management of their external impact.

The policy recommendations emerging from the analysis of Swedish cities action confirm that local efforts, while essential, cannot succeed in isolation. Effective urban measures to manage CBE and mitigate spillover effects require close, vertical collaboration with national policy frameworks. This national policy, in turn, must be consciously aligned with EU regulations and the broader global agenda. The challenge of preserving the planet for future generations is fundamentally one of multi-level governance. By creating coherent, interconnected policy levers spanning local, national, and international levels, we can transform the theoretical understanding of planetary interdependence into practical, positive solutions. Only through a

collective commitment to global ecological footprint management can we ensure that prosperity in one region does not come at the expense of environmental stability worldwide.

For future research several areas can be researched further to support the transition to globally responsible cities. A need for the development and testing of practical methodologies for municipalities should be research to quantify a broader range of non-carbon spillovers, particularly the embodied water, biodiversity, and social impacts in key consumption categories like construction, textiles, and electronics. As cities adopt new ways of measuring these impacts, further empirical research is needed to critically evaluate the long-term effectiveness of the "soft" governance models they currently rely on, such as voluntary contracts and multi-stakeholder collaboration; comparative studies could assess under what conditions these orchestration-based models succeed or fail in driving systemic change compared to more traditional regulatory instruments. This leads to the crucial question of public acceptance, meaning research should also explore the political feasibility and psychology of sustainable consumption, including the social and psychological barriers to behaviour change among high-impact groups and effective communication strategies for policies that directly target high-consumption lifestyles. Finally, to provide a viable long-term strategic direction for these efforts, future research can explore how theoretical concepts like Doughnut Economics or post-growth can be translated into practical and politically viable strategies for municipal planning and economic development, offering alternative pathways to prosperity that do not rely on ever-increasing material and energy throughput.

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8. Appendix

Table 9: The longlist consumption-oriented policies in the Nordic countries.

Measure (Engström et al., 2024)	Description (Engström et al., 2024)	Applicable	Reason
Product labels	Requirement to provide information on product labels on emissions of GHG and other air pollutants.	No	Demand product labels national (EU) level policy.
Negative labelling	Food products with negative labelling that signals if there is an environmental or health problem with the product.	No	Demand product labels national (EU) level policy.
Positive labelling	Ecolabelling or certification can reduce the environmental impact by encouraging producers to change production and by influencing consumer choices.	No	Demand product labels national (EU) level policy.
Environmental declaration	Forms of environmental declaration include energy declaration on the energy performance of houses but also graded labelling which combines negative and positive labelling.	No	Demand product labels national (EU) level policy.

Information campaign	Campaigns that inform consumers on products and behavior that cause high emissions of GHG and other air pollutants.	Yes	Directly applicable. Cities can run public service campaigns on sustainable consumption, waste reduction, and low-emission lifestyles.
Promotions for vacationers in the home country	Campaigns to increase the status of vacationers in the home country, such as cottage vacations, active vacations or learning vacations such as summer courses.	Yes	Cities can launch tourism campaigns promoting local "staycations" and nearby attractions to reduce emissions from long-distance travel.
Consumer guides and dietary advice	Mandatory information available to consumers on environmental and health effects of food items and diets.	Yes	Local health and environmental departments can create and distribute guides on sustainable food choices.
Environmental education	Develop pedagogical materials that can help teachers increase students' knowledge about food and its environmental impact.	Yes	Cities can develop and integrate educational materials on consumption and climate impact into local school curricula.
Energy and climate advice	A free service that helps consumers become more energy-smart and reduce their energy use, covering	Yes	Directly applicable. Cities can have local energy

	topics like heating, efficiency, transport, and costs.		advisory services to help residents reduce their consumption.
The hourly measurement reform	Gives electricity users incentives to change their consumption pattern to shift electricity use from high-price to low-price times.	No	This is national level policy.
Energy labelling	Shows how energy-demanding a product is on a scale from A to G, with arrows from green to red.	No	Demand energy labeling is national (EU) level policy.
Product location at point of sale	Low carbon products are given preferential placement at retail stores, internet sites, etc.	Yes	Cities can negotiate voluntary agreements with local retailers to give preferential placement to low-carbon products.
Rankings and award campaigns	Product manufacturers and/or sellers are given publicly celebrated awards for low carbon performance through government, trade or third-party organisations.	Yes	A city government can create its own awards to recognize and promote local businesses with low-carbon practices.
Approved technology lists	A list of specific low carbon technologies that are given preferential procurement.	Yes	A city's procurement department can create and use lists of approved low-carbon technologies for all municipal purchases.

Graduated tax on advertising	Size of VAT on advertising increases with increasing carbon implications, within each class of products.	No	This is a national-level fiscal policy.
Advertising regulation	Advertisement restrictions regarding where, when or how products or services with high emission intensities can be marketed.	Yes	Cities can create bylaws to restrict advertising for high-emission goods and services in specific zones, such as near schools or on public transport.
Business emission agreements/allowances	Businesses are required to acquire allowances for Scope 1 & 2 (at least) emissions, generally with trading.	No	Cities don't have the authority to set up a trading system for business emissions.
Requirements for improved sustainability	Successively increased demands are placed on food sold in the grocery store to be sustainable, for example through targets/requirements relating to climate, biological diversity, health and social aspects.	Yes	A city can mandate sustainability requirements for food served or sold in its own public institutions like schools, hospitals, and government cafeterias.
Public procurement	Requirements that public agencies must choose low or zero emission products or services when available.	Yes	Highly applicable. Cities are major purchasers of goods and services and can set strict requirements for low-

			emission products.
Reduce emissions from public employee business trips	Reduce the number of long-distance business trips by public employees in the state, regions and municipalities.	Yes	City government can implement policies to limit air travel for its employees and promote virtual meetings.
Linking public procurement to national climate change acts	Legal requirements to consider national climate goals in all public procurement.	Yes	Cities can align their procurement policies with national climate goals through local regulations.
Consumer carbon budget/personal carbon allowances	Consumers are provided an annual carbon budget and cannot exceed this, perhaps with allowance for trading.	No	This is a national-level policy that would require a complex system of tracking and enforcement, beyond the scope of a city.
Flight rights	Consumption is regulated by allocating emissions rights to consumers for flights, to reduce emissions from flying.	No	Cities cannot issue or regulate “flight rights”.
Consumption rights for meat	Regulate meat consumption by distributing consumption rights for meat to the consumer.	No	This is a national policy which is requiring a centralized system for tracking and trading consumption rights for a specific food type.

Regulatory standards	Regulation of emissions from products.	No	This is a national policy.
Reduction obligation for aviation fuels	Blending of biofuel into jet fuel, with timed quotas for the proportion of biofuel that must be mixed in time.	No	Regulating aviation fuels is a national or international policy responsibility.
Sector trade body standards	Voluntary product performance standards set by trade organisations to be followed by all outlets in that trade.	Yes	Cities can partner with local trade associations to encourage the adoption of voluntary standards for businesses within city limits.
Licenses	A license is required either to sell or purchase high carbon products.	No	Licensing is a national-level regulatory power, especially for products with carbon implications.
Supply chain procurement requirements	Consumer-facing outlets establish embodied carbon requirements on intermediate producers.	Yes	A city can require its contractors and suppliers to meet specific embodied carbon standards, influencing the supply chain.
Voluntary agreements by trade organisations	Trade organisations adopt voluntary commitments to reducing embodied and/or usage carbon of products offered to consumers.	Yes	Directly applicable. Cities can work with local trade organizations to establish voluntary commitments

			for their members.
Required recycling	A ban on disposal of products and materials that can be reused or recycled is introduced.	No, but partially yes	This is national level legislation. However, cities play a key role in enforcing waste management regulations and may require mandatory recycling of certain materials within national legislation
Sale of leftover food	Offer a range of things based on leftover food from restaurants or commercial kitchens.	Yes	Directly applicable. Cities can create policies, partnerships, or local marketplaces to facilitate the sale of leftover food from restaurants.
Extended producer responsibility	Producers have responsibility for collecting and recycling products at end of use.	No	This is a national-level policy, which requires a legal framework to enforce responsibility for a product throughout its lifecycle.
Product ban	Products are banned based on criteria of embodied and/or usage carbon.	No	Banning products is a national regulatory policy.
Prohibition of certain vehicles in urban zones	An environmental zone limits the type of vehicle	Yes	Directly applicable.

	that may be driven within a specific area.		Many cities have already implemented this, such as low-emission zones or congestion charging zones.
Ban on short flights	A short-haul flight ban is a prohibition imposed by governments on airlines to establish flight connections over short distances.	No	Regulating flight routes and distances is a national or international policy area.
Shop product choice	Point of sale operators voluntarily restrict products to lower embodied and/or usage products.	Yes	Cities can promote voluntary agreements with local retailers to restrict high-carbon products on their shelves.
Waste targets, requirements and/or prices	Product recycling is motivated through waste policies that place either a requirement for, or a price on waste generation.	No	This is national level policy.
Deposits on purchased goods	Deposits are initiated to enhance recycling of goods to reduce raw materials requirements.	No	This is national level measure.
Repair checks and repair funds	Consumers get access to repair vouchers when purchasing certain product groups which can be redeemed to reduce the price of repairs or upgrades to the product.	Yes	Cities can create local repair voucher programs or partner with businesses to offer repair services at a reduced cost.
Minimum price limits	Very low prices are banned to remove from markets products that have less	No	This is a national economic policy. Cities do

	incorporation of externalities.		not have the authority to set minimum prices on goods.
Limits on percentage ownership or use	Nations or municipalities restrict the number of a given product (such as cars) that can be purchased and/or owned.	No	This is a national-level policy, as it requires extensive tracking of individual ownership.
Extension of product lifetime	Enhancement of product lifetime, including removal of planned obsolescence, and ban on disposal of unsold and functional products.	No, but partially yes	Demanding and banning products is national level policy. However, cities can promote and support local repair services, community workshops, and second-hand shops to extend product life.
Ban on disposal of unused products	Companies are prohibited from disposing of unsold and functional products and are required to instead resell, donate or reuse products.	No	This is a national level policy.
Information on service life and repairability	Introduce requirements for labelling with expected lifespan of goods within durable goods, linked to a lifespan guarantee.	Yes	Cities can promote transparency and educate consumers about product durability and repair options, although labeling is a national level policy.

Minimum requirements for repairability	Requires manufacturers to design products to be taken apart and repaired, and to provide spare parts.	No	Setting minimum requirements for product design is a national or EU regulatory function.
Extended right of complaint	Extend the statutory period for right of complaint, which would give companies an incentive to sell goods with better quality.	No	This is a national consumer protection law that cannot be set by a city.
EU Eco-design for Sustainable Products Regulation	A framework to set eco-design requirements for specific product groups to significantly improve their circularity and environmental sustainability.	No	This is a national level policy.
Carbon tax	A tax levied on most fossil fuels in proportion to their carbon content.	No	Levying a carbon tax is a national fiscal policy.
Fuel and vehicle taxes	A key policy instrument for reducing CBE.	No	Levying these taxes is a national or state-level policy.
Carbon Border Adjustment Mechanism	Importers bringing certain types of goods into the EU from third countries must declare embedded emissions and purchase corresponding certificates.	No	This is an international trade policy.
Consumption taxes	Many countries have consumption taxes based on emissions of GHG on petrol and diesel. This could be expanded to other product categories.	No	This is a national fiscal policy.
Excise duties	Introduce point taxes to reduce or redirect consumption from	No	This is a national fiscal policy.

	environmentally harmful or unhealthy goods and services.		
Flight tax	A tax on air travel tickets.	No	This is a national tax on air travel.
Frequent flyer tax	A progressive tax on air travel, with an increasing tax the more you fly.	No	This is a national-level tax.
Taxes on unsustainable food	Taxes on meat and dairy products so that the price for the consumer is increased to reduce the consumption of these and thus also the climate impact.	No	Levying a tax on food is a national fiscal policy.
Bonus-malus systems	A system where products with relatively low emissions of carbon dioxide are rewarded at the time of purchase with a bonus, while products with relatively high emissions of carbon dioxide are charged with a higher tax.	No	This is a national fiscal policy.
Bonus-malus system for flights and trains	Introduce a system with an integrated air-train tax, where increased taxes on air travel are used to direct subsidies to train travel.	No	This is a national fiscal policy.
Bonus-Malus system for home decoration	Products with a low environmental impact and a long lifespan are stimulated by a lower fee, while goods with a higher environmental impact and a shorter lifespan are charged a higher fee.	No	This is a national fiscal policy.
Cash compensatory measures	Targeted, compensatory measures proposed to address the regressive nature of environmental taxes (i.e., affecting low-income people more).	Yes	Cities can implement local programs or work with national programs to provide

			targeted financial assistance to low-income residents affected by environmental taxes.
Fee & dividend for climate-harming consumption	Environmental taxes are combined with a refund targeted specifically towards low-income households.	No	This is a national fiscal policy.
Subsidy	Subsidies directed at purchases of low emission products which have the potential to grow and push out technologies or practices with higher emissions.	No	This is a national fiscal policy.
VAT differentiation	Regulate the VAT so that it is, for example, lower on services and used goods, which would benefit, for example, repairs over new purchases.	No	This is a national fiscal policy.
Subsidize repair services	Subsidize repair services to increase the profitability of these companies/organizations and, by extension, speed development.	No	This is a national fiscal policy.
Investment subsidies	Investment subsidies which reduces demand for high carbon products, for instance subsidies for electric vehicles or home batteries for solar power.	No	This is a national fiscal policy.
Industry subsidies	A key policy instrument but outside the explicit scope of this mapping.	No	Subsidies for entire industries are a national-level policy.

Remove climate-damaging subsidies for aviation	For example, municipalities and regions that currently support, or own local airports can remove subsidies and deny new construction or expansion of runways.	Yes	Cities can end their financial support or ownership of local airports to reduce aviation's carbon footprint.
Abolish animal subsidies	Abolish subsidies for animal consumption, such as the "school milk allowance".	No	This is a national agricultural policy.
Subsidies on sustainable food	Subsidize products and thereby lower the price for the consumer to get a consumer to choose a product with less climate impact than one with a higher impact.	No	This is a national fiscal policy.
Tax deductions for sustainable services/products	Tax deductions can for instance, be given to companies/organisations which work with upcycling products.	No	This is a national tax policy.
Subsidize second-hand sales	Subsidize second-hand sales to increase the profitability of these companies/organizations and, by extension, speed up development.	No	This is a national fiscal policy.
Energy efficiency	Measures directed at lowering electricity or other power demand from private households.	No	This is national level policy.
Tax reductions for home installation of green technology	Tax reductions for home installation of green technology	No	This is a national tax policy.
Support for certain environmentally-enhancing installations in single-family houses	Support for certain environmentally-enhancing installations in single-family houses.	No	This is national level policy.
Support conversion from oil heating in residential buildings	Support conversion from oil heating in residential buildings.	No	This is national level policy.

Investment support solar cells	Investment support solar cells.	No	This is national level policy.
Product user fees	A fee is attached at point of sale based on carbon associated with subsequent use.	No	This is a national-level policy, as it would be difficult to track and enforce at the city level.
Refund mechanism	Part of the price of purchase is refunded based on the product re-entering the recycling stream.	No	This is a national fiscal policy.
Preferential finance terms	Finance terms on loans – including credit cards – tied to carbon implications of product choice within a given category of goods.	No	Cities cannot regulate the interest rates or finance terms
Finance policy	A key policy instrument but outside the explicit scope of this mapping.	No	This is a national-level policy.
EUs sustainable finance taxonomy	A key policy instrument but outside the explicit scope of this mapping.	Yes	Cities can adopt this taxonomy to guide their own investments and procurement decisions.
Enabling recycling	Facilitate for consumers to re-cycle products and materials at end of use.	Yes	Directly applicable. Cities are responsible for providing recycling infrastructure.
Enabling product sharing	Creation of the infrastructure to enhance the ability of consumers to share products rather than purchase individual items.	Yes	Directly applicable. Cities can support local sharing platforms and libraries of things.

Spaces for sharing	Requirement that developers set aside space for the sharing economy.	Yes	Directly applicable. Urban planners can require new developments to include spaces for sharing.
Mandatory metering	Requirement of metering for at least heat and electricity in buildings, and potentially for automobile use.	No	This is national level policy.
Infrastructure improvements	A key policy instrument but outside the explicit scope of this mapping.	Yes	Directly applicable. Cities are responsible for planning and building infrastructure like bike lanes and public transit.
R&D and innovation policy	A key policy instrument but outside the explicit scope of this mapping.	Yes	Cities can fund local R&D in areas like green technology and sustainable urban design.
Work time reduction	Employers and other organizations can choose to facilitate sustainable lifestyles by offering the opportunity to reduce working hours (with a corresponding reduction in wages).	No, but partially yes	This is a company-level or national policy, but it can be applied to city employees.
Targets and accounting of CBE	Set targets for CBE where emissions outside the country's borders are also included.	Yes	Directly applicable. Cities can set their own CBE targets and track their progress.

Climate calculations for consumers	Tools that calculate individual climate impacts from transport, housing, consumption and food and provide inspiration for how you can live more sustainably.	Yes	A city can develop and promote tools or apps for residents to calculate their own carbon footprint.
Consumption-based Paris-compliant 1.5 carbon budgets as a climate policy framework	Calculated for the European Union.	Yes	Cities can adopt the principles of the framework.
International cooperation	A key policy instrument but outside the explicit scope of this mapping. High-income countries can direct investment and technology transfer to developing countries for their decarbonization.	No	This is a national policy.
The Green Climate Fund	A key policy instrument but outside the explicit scope of this mapping. The UN Green Climate Fund is currently the world's largest multilateral fund to help developing countries finance climate mitigation.	No	This is an international fund and is not within a city's competence.
Tax free shopping	Norway Customs Regulations. Tax free - possibility to bring in goods duty free for personal use (encourages flying rather than the opposite).	No	This is a national or international customs policy.
Conditioning tax deduction for home repair	Tax deductions can be made for costs for home-repair if the work increases energy efficiency or reduces overall use of resources and materials.	No	This is a national tax policy.
Reduce incentives for commuting to work by car	Tax deductions for expenses from commuting to work should be neutral in relation to way of transport.	No	This is a national tax policy, though cities can influence it

			through lobbying.